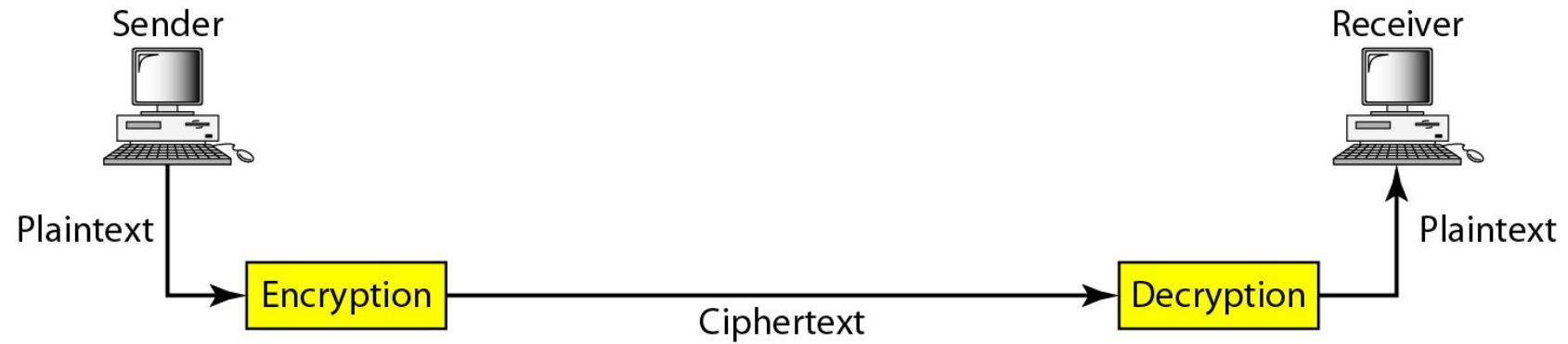

Figure 30.1 *Cryptography components*



Basic Terminology

- **plaintext** - the original message
- **ciphertext** - the coded message
- **cipher** - algorithm for transforming plaintext to ciphertext
- **key** - info used in cipher known only to sender/receiver
- **encipher (encrypt)** - converting plaintext to ciphertext
- **decipher (decrypt)** - recovering ciphertext from plaintext
- **cryptography** - study of encryption principles/methods
- **cryptanalysis (codebreaking)** - the study of principles/ methods of deciphering ciphertext *without* knowing key
- **cryptology** - the field of both cryptography and cryptanalysis

Conventional Encryption Principles

- An encryption scheme has five ingredients:
 - Plaintext
 - Encryption algorithm
 - Secret Key
 - Ciphertext
 - Decryption algorithm
- Security depends on the secrecy of the key, not the secrecy of the algorithm

Characteristics of Cryptographic Techniques

- Classified along three independent dimensions:
 - The type of operations used for transforming plaintext to ciphertext
 - The number of keys used
 - symmetric (single key)
 - asymmetric (two-keys, or public-key encryption)
 - The way in which the plaintext is processed

Classical Substitution Ciphers

- where letters of plaintext are replaced by other letters or by numbers or symbols
- or if plaintext is viewed as a sequence of bits, then substitution involves replacing plaintext bit patterns with ciphertext bit patterns

Caesar Cipher

- earliest known substitution cipher
- by Julius Caesar
- first attested use in military affairs
- replaces each letter by 3rd letter after
- example:

meet me after the toga party
PHHW PH DIWHU WKH WRJD SDUWB

Caesar Cipher

- can define transformation as:

a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C

- mathematically give each letter a number

a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25

- then have Caesar cipher as:

$$c = E(p) = (p + k) \bmod (26)$$

$$p = D(c) = (c - k) \bmod (26)$$

Monoalphabetic Cipher

- rather than just shifting the alphabet
- could shuffle (jumble) the letters arbitrarily
- each plaintext letter maps to a different random ciphertext letter
- hence key is 26 letters long

Plain: abcdefghijklmnopqrstuvwxyz

Cipher: DKVQFIBJWPESCXHTMYAUOLRGZN

Plaintext: ifwewishtoreplaceletters

Ciphertext: WIRFRWAJUHYFTSDVFSFUUFYA

Polyalphabetic Ciphers

- **polyalphabetic substitution ciphers**
- A set of related monoalphabetic substitution rules is used
- use a key to select which alphabet is used for each letter of the message
- use each alphabet in turn
- repeat from start after end of key is reached
- make cryptanalysis harder with more alphabets to guess and flatter frequency distribution

Key: deceptive 3 4 2 4 15 19 8 21 4

plaintext: wireless 22 8 17 4 11 4 18 18

ciphertext: zmtiaxao 25 12 19 8 26 23 26 39

Vernam Cipher and One-time Pad

- Keyword is as long as the plaintext and has no statistical relationship to it.
- Vernam system works on binary data with ith bit of text exclusive or'd with ith bit of key to produce ith bit of cipher
- In one one-time pad key is used only once
- This scheme is unbreakable

